# Language Agent Tree Search

One frozen model as a planning agent

Weekend 6 · Frontier Agentic AI

Carlos Cotrini

# Where LATS sits

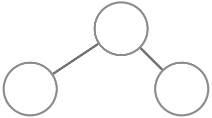
| | | | |
|---|---|---|---|
| agents | **LATS** | all of them, inside a search | **we are here** |
| | Monte Carlo Tree Search | look ahead before moving | this morning |
| | ReAct | think, act, observe | |
| prompting | Tree-of-Thought | branch, score, keep the best | |
| | Chain-of-Thought | write the reasoning first | |
| the model | Reasoning model | trained by RL to reason before answering | |
| | Large language model | a transformer compresses any prefix | |
| | Language model | predict the next word | |

Section 1

# The LATS method

1

# The same loop as this morning, with four edits



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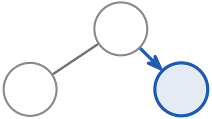

**1 Selection** walk down by UCT

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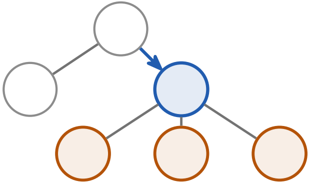

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| **1 Selection** | walk down by UCT |
| **2 Expansion** | the *actor* proposes candidates |

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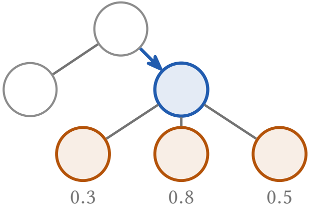


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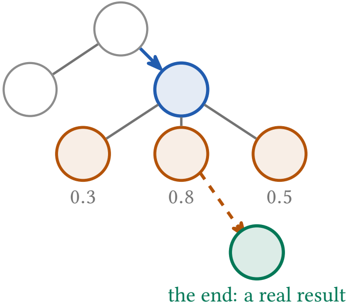


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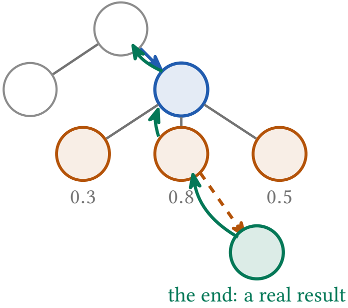


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| **5 Backpropagation** | push the *real* result up |

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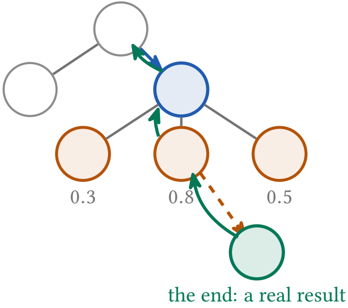


a note on what went wrong

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**2 Expansion** — the *actor* proposes candidates

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**6 Reflection** — the *feedback* model writes it

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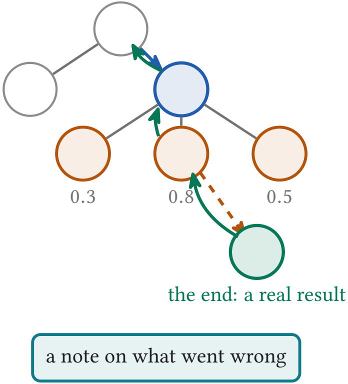


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amber: what changed since this morning

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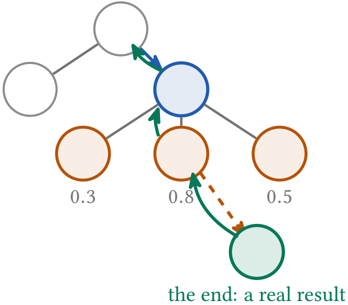


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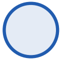
into the next prompt

amber: what changed since this morning

# What the critic actually returns

a state 

$$V(s) \;=\; \qquad\qquad +$$

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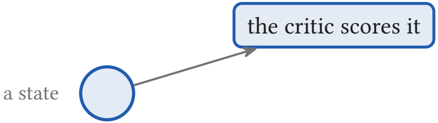


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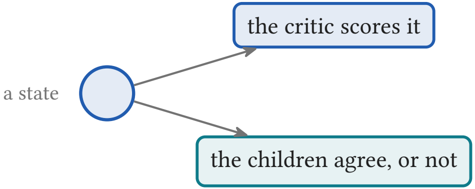


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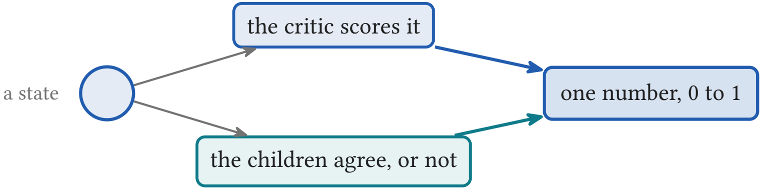


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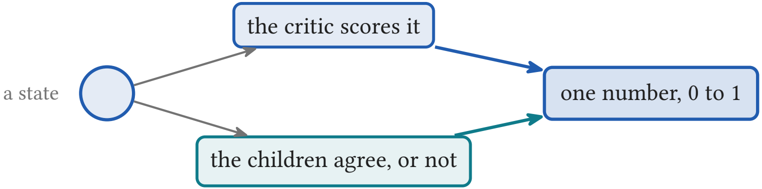


$$V(s) \;=\; \lambda\,\mathrm{LM}(s) \;+$$

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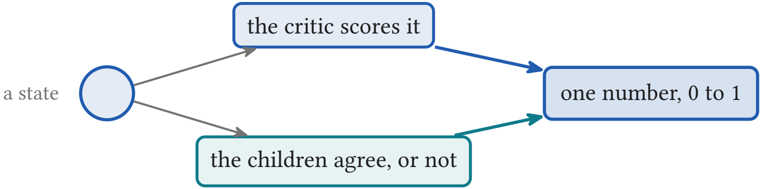


$$V(s) = \lambda\,\mathrm{LM}(s) + (1-\lambda)\,\mathrm{SC}(s)$$

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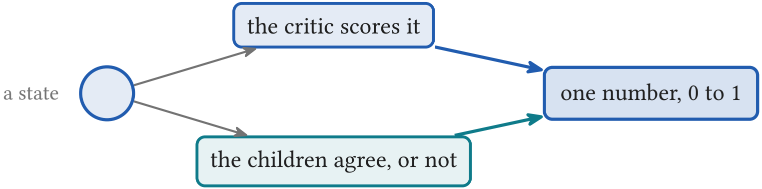


$$V(s) = \lambda\,\mathrm{LM}(s) + (1-\lambda)\,\mathrm{SC}(s)$$

$\lambda = 0.5$ on question answering, 0.8 on programming and web tasks.

# Three jobs, one set of weights

one frozen model

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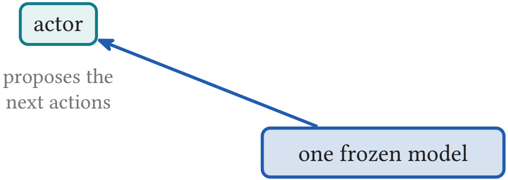

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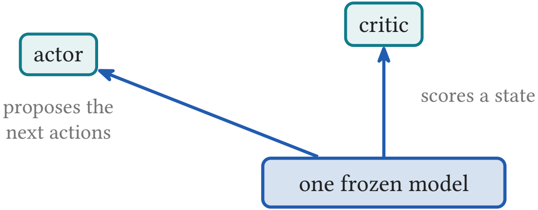

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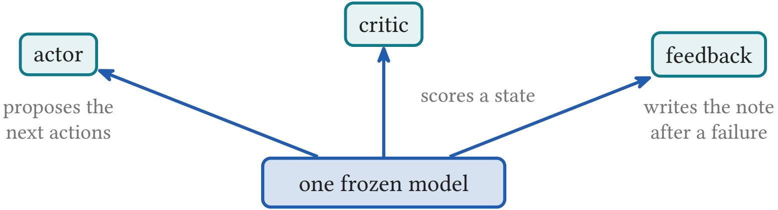

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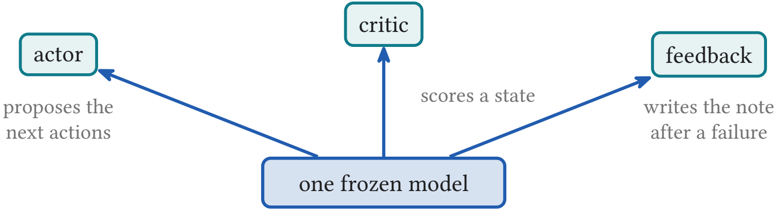
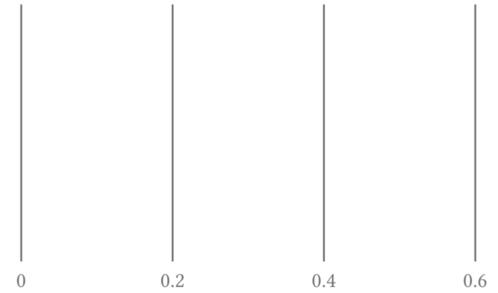



same weights every time. Only the prompt changes.

# What happens when you remove each piece



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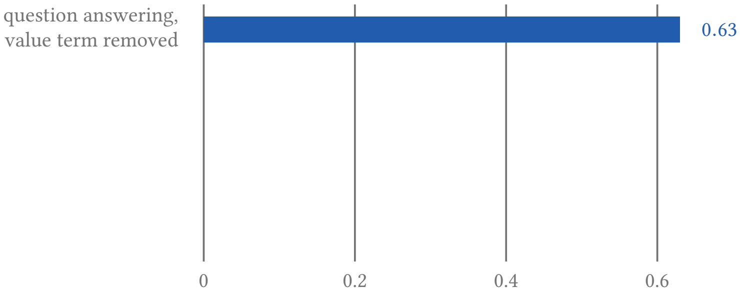

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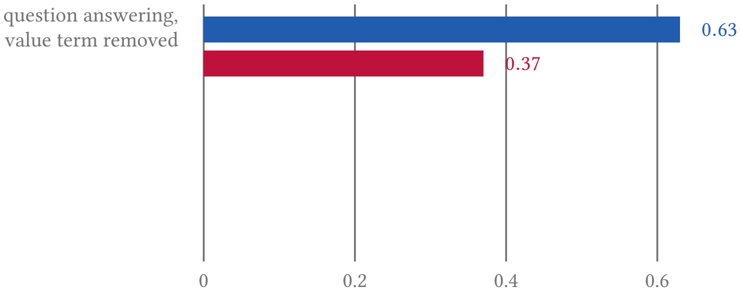

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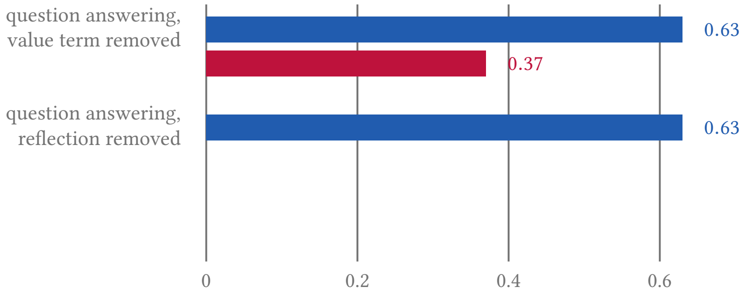

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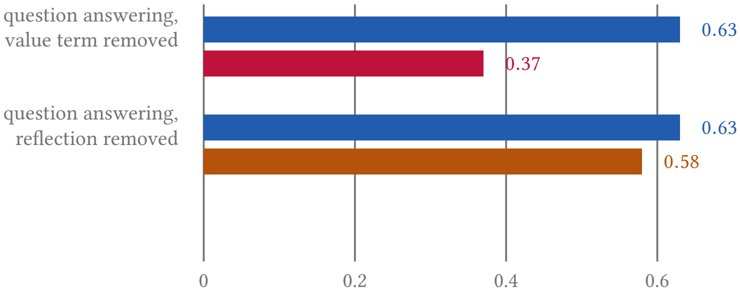

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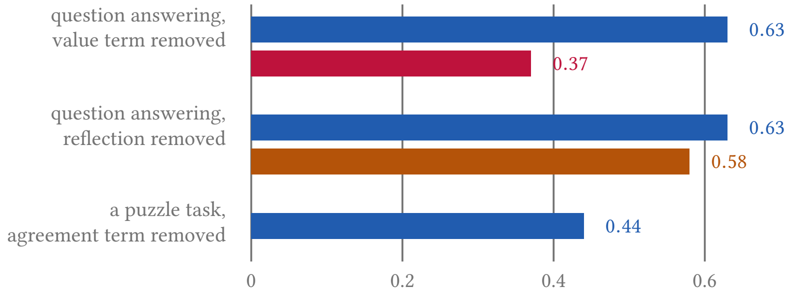

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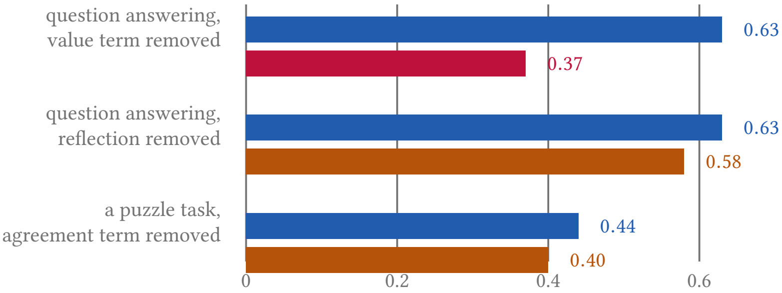

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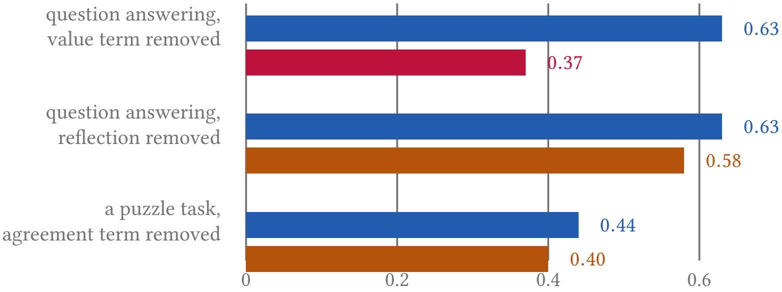


The critic's score is doing almost all of the work. The last row is a different task, so read it on its own.